Background

In complex systems, there is almost never one study or piece of evidence that conclusively proves one hypothesis is true. Every study has limitations, every source has biases, every population has outliers, and every statistic has confounders. It is therefore necessary to integrate all evidence into a coherent picture.

Following is the hypothesis we found best explains all evidence relating to vaccine effectiveness:

1. Both mRNA vaccines induce a robust immune response that makes the immune system more effective in fighting the virus.
2. For a few months immunity is effective against symptomatic infection.
3. As the response subsides and retreats to immune memory, it takes longer to mount an effective response, causing $VE_{infection}$ to wane from >90% to negligible.
4. Since death normally takes weeks from infection, this process has minimal effect on VE_{death}, which is maintained at 90% for years.
5. New variants, especially since omicron, bypass protection from infection, but are mostly still susceptible to the long-term immune response that determines VE_{death}.
6. Boosters trigger a new immune response, thus recovering $VE_{infection}$ for a few months, but as VE_{death} is already high, they can provide only modest improvements.
7. Over time, and especially late 2021, the unvaccinated got infected, and more frail unvaccinated died, thereby closing much of the susceptibility gap between the two groups, creating a reduction in hybrid VE_{death}, but not in the real VE_{death}.

We will now examine how all evidence fits well with that hypothesis, emphasizing how each type of evidence has different weaknesses, thus generating high confidence in the hypothesis.

RCTs

We have two independent RCTs that produced near identical results (Moderna, Pfizer). While not powered for deaths they provided good data on preventing severe Covid:
For Pfizer the ratio was 9:1, and for Moderna 30:0. This is a VE of 97.5% (95% CI, 81.3 to 99.7).
It is reasonable to assume (and agreed by the opposition) that VE_{death} would be similar or higher to VE for severe Covid.

Strengths:

- No selection bias, HVE, hypothesized NPH, etc. (other than described below)
- Prior infections were verified and excluded, producing reliable VE estimates

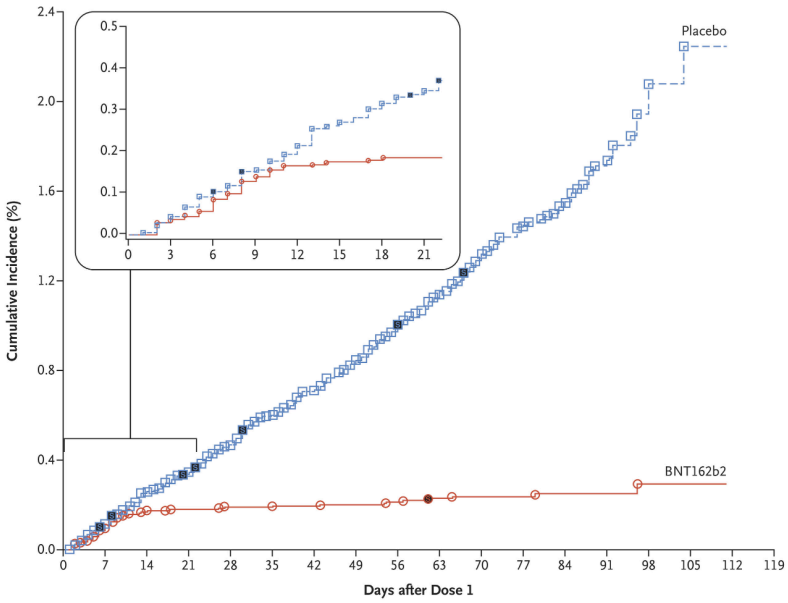
LImitations:

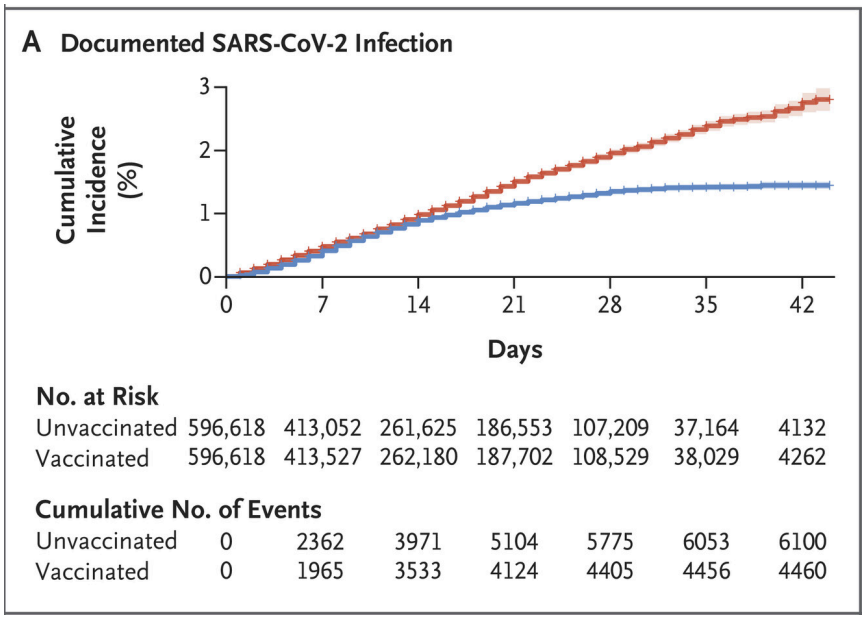
- Subjects were relatively young and healthy, so not representative of the population that actually accounts for most lives saved.
- The timeframe was a few months, meaning the RCTs provide little information on waning.

- While the studies were large in absolute number, they're still a negligible part of the population, so provide no information on how widespread vaccination could select for new variants.
- Unblinding is relatively easy due to common vaccine side effects. However, for the severe Covid endpoint that should be a minimal bias (mostly determined by objective markers measured at a hospital).

Replicating RCT results

The first observational study was from Israel and it replicated the RCT results nearly perfectly, including near identical Kaplan-Meyer curves (Pfizer RCT on the left, the observational study on the right), and 92% (95% CI, 75 to 100) reduction of severe Covid. The population was over 1 million Israelis vaccinated prior to 1-Feb-2021, meaning these were some of the first to vaccinate in the world, and biased towards elderly.





This addresses several major concerns:

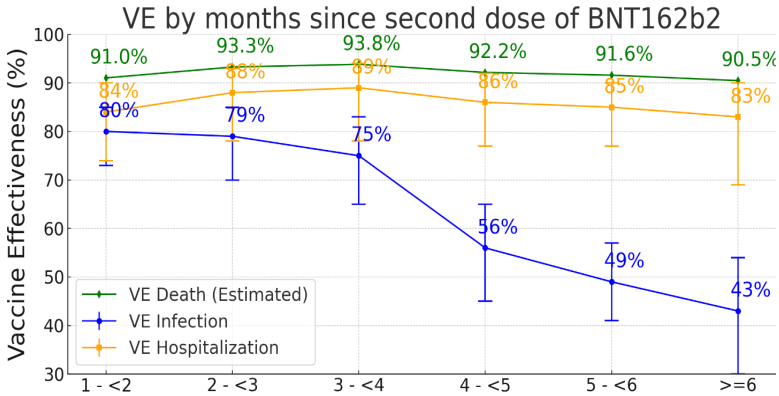
- The healthy population of the RCTs is reasonably representative of the target population.
- Unblinding is not a major bias.
- Observational studies are able to produce reliable results despite HVE and other selection biases.

Concerns about variants and waning still apply.

Note that this alone is enough to conclude net benefit. We provided in V2 (“Absolute Minimum”) a calculation of lives saved assuming the vaccines become completely useless after 4 months, reaching 60,000 lives saved. This still far exceeds any reliable estimate of VID.

Observational Studies

Once we see observational studies can be fairly accurate in estimating effect on Covid deaths, the hundreds of VE studies showing high effectiveness become reliable.
We refer to multiple studies here, including RCTs, test-control negative studies, cohort studies, and others (note that each type of study has different confounders!), all showing high vaccine efficacy. Some of them claim waning VE_{death}, but that is only for hybrid VE, and at a period when unvaccinated have gained substantial natural immunity, closing the immunity gap between the groups.
Therefore, when asked for "a good VE study" (again, emphasizing that complex issues cannot be settled with a single study) we often give the Kaiser Permanente study (chart below), which did control for NI, and indeed doesn't show waning VE_{death}



Selection Bias

The main claim of the opposition is that all findings of lower death rate in vaccinated are due to selection bias, specifically HVE, with a further claim that HVE is higher for Covid deaths than for ACM ("NPH").
First, we note that while HVE is seen in ACM, we have not found any study claiming to find HVE in Covid deaths. There are actually good reasons to hypothesize it would be lower there. While it makes sense not to vaccinate a dying cancer patient, or a patient with advanced dementia who's not dying soon but has very low quality of life, there is no reason not to vaccinate someone who is reasonably healthy but more susceptible to Covid, such as someone with low lung capacity who interacts with many people. Governments were universally pushing to vaccinate these individuals early and more frequently. This is not to say people who didn't vaccinate are not also likely to be more careless in other aspects related to Covid risk, but overall, HVE for Covid deaths, if it even exists, should be lower than for ACM, not higher.
Beyond this claim being unsubstantiated, there is ample evidence that VE=90% is a true finding. Following is a list of conditions under which it has been observed, each one having **different** biases, making it highly unlikely that all findings are false.

Consistent among different geographies and vaccination rates

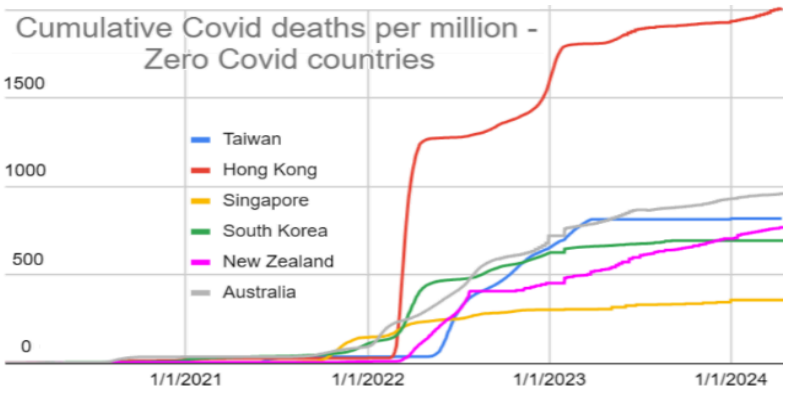
The VE studies were replicated in all geographies, despite each having different vaccination

policies.
Most notable was Hong Kong which had a low vaccination rate for the elderly. Selection bias is generally much lower when the groups are of similar size (it's easy to bias a 5% unvaccinated group with dying patients, but much harder with 50%). The age groups closest to 50% in Hong Kong were 65+ (48% unvaccinated) and 70-79 (52%). Their VE_{death} was 95% and 93%, respectively.
Note: While most were mRNA at the time, SinoVac was also popular.

VE maintained after one year of zero Covid

The same study also provides further evidence that the decline in VE is due to unvaccinated gaining natural immunity and not real waning, since it shows VE>90%, a year after vaccinations, following a successful zero Covid policy (i.e. no NI).
This is further supported by data from another zero Covid geography (New South Wales, Australia - see V2 "NSW"), shared by Judge N. While the data showed that over time vaccinated accounted for 90% of Covid deaths, this is the expected result given VE=90% and a 99% vaccination rate of elderly.

Hong Kong's low vaccination rate is also evident when comparing their deaths to the other zero covid countries:



Applies also to changes in ACM, relative to low-Covid baseline

We analyzed the Czech dataset (V5 "ACM by Vaccination Status"), and showed that the increase in ACM of the unvaccinated was 10 times the increase in the vaccinated (relative to the baseline trend in ACM during low Covid periods).

- This directly addresses selection bias concerns, as the increase was measured only relative to the baseline trend.
- The same pattern repeats in ACM data from UK and NL, despite different baseline death rates (i.e. different selection bias)
- There is still the hypothesized possibility of a selection bias that is highly specific to Covid deaths, but it is highly unlikely for such a bias to account for a 10x difference (on top of the existing bias during low Covid periods), and replicate in different countries with different vaccination policies.

Consistent among different age groups, including when selection bias seems negligible

We further reviewed the UK ACM data (V5 “United Kingdom”), which is especially interesting as it provides breakdown by age, and shows that each age group has different baseline trends, indicating different HVE or selection bias. The 40-49 age group is most interesting, as they show negligible difference in baseline ACM between the groups, indicating low selection bias, and still they have the same pattern of unvaccinated ACM increasing during Covid, while vaccinated are hardly affected.

Natural immunity reduces hybrid VE, indicating it is real

In V5 (“Verifying VE from Share of Unvaccinated Deaths”) we’ve shown that the share of vaccinated among Covid deaths matches perfectly VE=90%, with hybrid VE gradually declining towards 50%.

- While this data has confounders that are similar to other population level metrics, the fact that VE declines following Covid waves is evidence that this VE is real. If VE was just a result of selection bias, then the spread of natural immunity would affect both groups equally, keeping the artificial VE constant over time. The reduction indicates that the vaccine does provide protection that is comparable to natural immunity (which no one denies is protective).

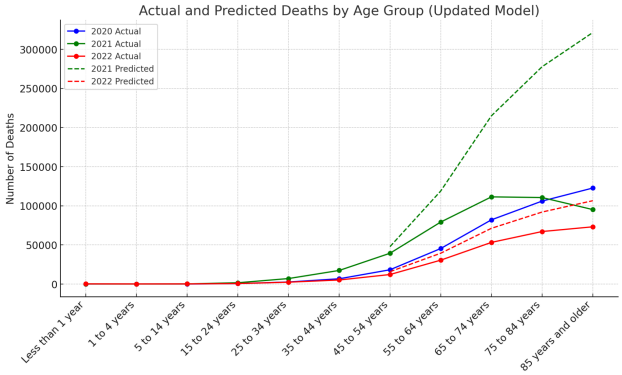
Vaccine protects from death, also when measuring only after hospitalization

- This study measured VE among patients **already admitted to hospital** in preventing progression to invasive mechanical ventilation or death and found it to be 76% for the alpha variant, 44% delta, and 46% omicron.
- This matches well the recent study submitted by the opposition, reporting 1.7x reduction from hospitalization to death a further one year later (i.e. lower hybrid VE).
- There is still the possibility of biases in the decision to hospitalize, but arguments can be made for both sides:
    - People who don’t vaccinate may be less likely to seek healthcare in general.
    - Unvaccinated may be hospitalized earlier as their doctors would perceive their risk of deterioration to be higher.
- Overall, this is a much less biased sample than the general population.
- Note that even if we assume the vaccine has no effect prior to hospitalization and only accept this 44-76% as reliable, it still translates to over 500,000 lives saved, as lives saved is mostly a linear function of VE.

Distribution of deaths by age group changed following vaccination

Following vaccinations we see a marked change in the distribution of Covid deaths by age, mostly in 2021, which is the most important as that is when most deaths and lives saved occurred. In 2022 the distribution was still less skewed towards old ages than in 2020, but less so than in 2021. This could be due to natural immunity gained by the lower-aged unvaccinated or due to omicron hardly affecting the young.
Since this focuses on age groups rather than vaccination status, there is no selection bias (people can’t choose their age). There could still be other confounders such as changes in

behavior by age group, or effect of variants



This study eliminates some of these residual confounders, as it examined the timing of the drop in deaths during Israel's vaccination campaign, and found they dropped earlier among older people, who were vaccinated first (this pattern was not seen in the preceding wave). Since this comparison is done within one single wave, differences in behavior and variants are minor.

Table of Evidence

Following is a summary of the sources above, demonstrating how their different limitations build high confidence in VE_{death}=90%

Source	Finding	Risks & Limitations Averted					
		HVE	NPH	Prior Infection / NI	Representation of high risk pop.	Waning	Variants
RCTs	VE_{severe} 97.5%	Randomized. Unblinding probably minimal		Tested & excluded	Young & Healthy	Too short	Ancestral
First observational (Israel)	VE_{severe} 92%	Not eliminated but results match RCTs which did		Tested & excluded	Elderly over represented	42d	Ancestral & Alpha
Kaiser Permanente	VE_{severe} ~85%	Controlled for healthcare-seeking behavior, SVI & pre-existing conditions		Tested & excluded	Elderly may be over represented	Controlled for & no VE_{death} waning	Alpha & Delta
Hong Kong	VE_{death} 90%	Not controlled for, but bias should be weaker after 1 year		After zero Covid		1 year and no waning observed	Omicron
CZ ACM Analysis	ACM increase during Covid ~10x in unvaxxed,	VE is measured relative to baseline trend			General population		All
UK age 40-49		No evidence of difference between the			Intentionally		All

ACM	matching VE_{death} 90%	groups		focusing on 40-49		
Age distribution of Covid deaths	vaccination rate associated with lower mortality, matching VE_{death} 90%	Measures entire age groups, and people can't choose their age		N/A		All
Timing of Covid death reduction	Vaccination rate associated with earlier reduction		Little NI at the time	N/A	Immediately after vaccination	Ancestral and Alpha
Hospitalization to death	In hospital VE_{death} 44-76%	Outcomes are less dependent on individual choices				Omicron, Delta, and Alpha

Summary

When attacking an hypothesis, it is not enough to point to limitations of specific sources in isolation, or to make blanket claims of selection bias. A complete coherent explanation must be provided that accounts for all evidence. We have shown that the evidence for high efficacy of Covid mRNA vaccines is vast and diverse, and each source has different non-overlapping limitations. Explaining all of them together would require an extreme string of coincidences whereby all confounders always work to support the net harm hypothesis, and exactly to the extent needed to artificially reach VE=90%. This is why neither the opposition, nor anyone else in the world, are able to produce a coherent reasonable net harm hypothesis that accounts for all the evidence.